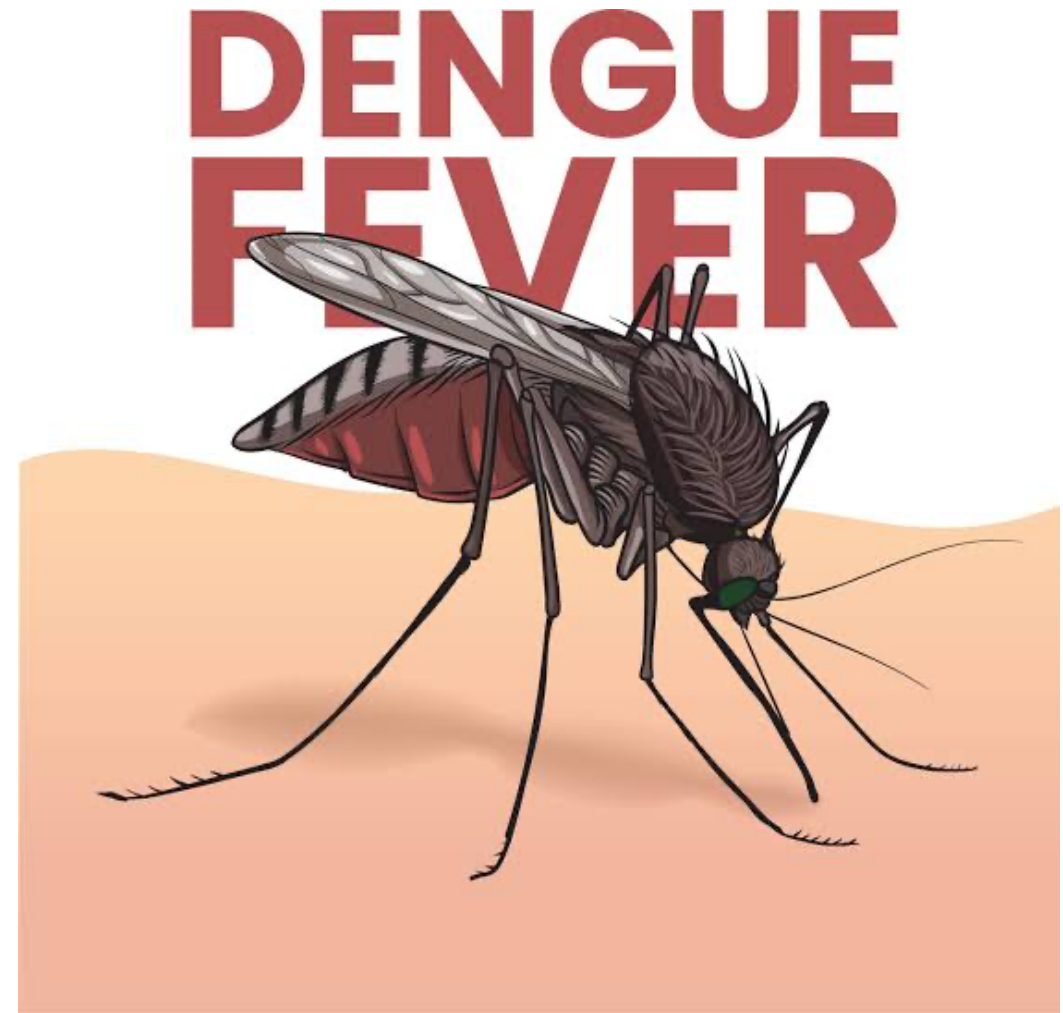


DengAI: predicting weekly dengue cases

Goal: estimate reported dengue cases each week in San Juan and Iquitos using environmental measurements

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3. Aibota Sanatbyek



The problem: dengue outbreaks are hard to anticipate



Dengue is a mosquito-borne virus spread by mosquitoes in tropical and subtropical regions. Severe cases can be fatal.

Outbreaks follow the weather. Warmth, humidity and rainfall drive mosquito breeding, so case counts rise and fall with the climate, often weeks later.

Why it matters

Earlier warning: predict a surge before it shows up in clinics

Lower cost: target spraying and breeding-site clean-up where and when it is needed

Better care: prepare beds, staff and test kits ahead of peaks

Our ML model: learn the link between environmental conditions and case counts, then forecast weekly cases for each city.

A supervised regression problem

We predict weekly dengue case counts for San Juan (Puerto Rico) and Iquitos (Peru).

```
In data.shape, labels.shape, test.shape
```

```
Out ((1456, 24), (1456, 4), (416, 24))
```

1,456 historical weeks, 416 future weeks, and 24 input columns.

```
In labels.head()
```

```
Out
```

	city	year	weekofyear	total_cases
0	sj	1990	18	4
1	sj	1990	19	5
2	sj	1990	20	4
3	sj	1990	21	3
4	sj	1990	22	6

total_cases is the value to predict:
a weekly count of reported cases.

For example, the first San Juan week has
4 cases.

The environmental features come from four sources

Group	Column names
Location and time	city, year, weekofyear, week_start_date
1. Weather stations	station_avg_temp_c, station_min_temp_c, station_max_temp_c, station_diur_temp_rng_c, station_precip_mm
2. Satellite rainfall	precipitation_amt_mm
3. Climate reanalysis	reanalysis_air_temp_k, reanalysis_avg_temp_k, reanalysis_min_air_temp_k, reanalysis_max_air_temp_k, reanalysis_dew_point_temp_k, reanalysis_tdtr_k, reanalysis_relative_humidity_percent, reanalysis_specific_humidity_g_per_kg, reanalysis_precip_amt_kg_per_m2, reanalysis_sat_precip_amt_mm
4. Satellite vegetation	ndvi_ne (northeast of the city centre) ndvi_nw (northwest), ndvi_se (southeast), ndvi_sw (southwest)

[Source: DrivenData feature definitions](#)

Check the cities and their timelines

San Juan (`sj`) and Iquitos (`iq`) have different amounts of history.

In

```
data.groupby("city")["week_start_date"]  
    .agg(["count", "min", "max"])
```

Out

city	count	min	max
iq	520	2000-07-01	2010-06-25
sj	936	1990-04-30	2008-04-22

In

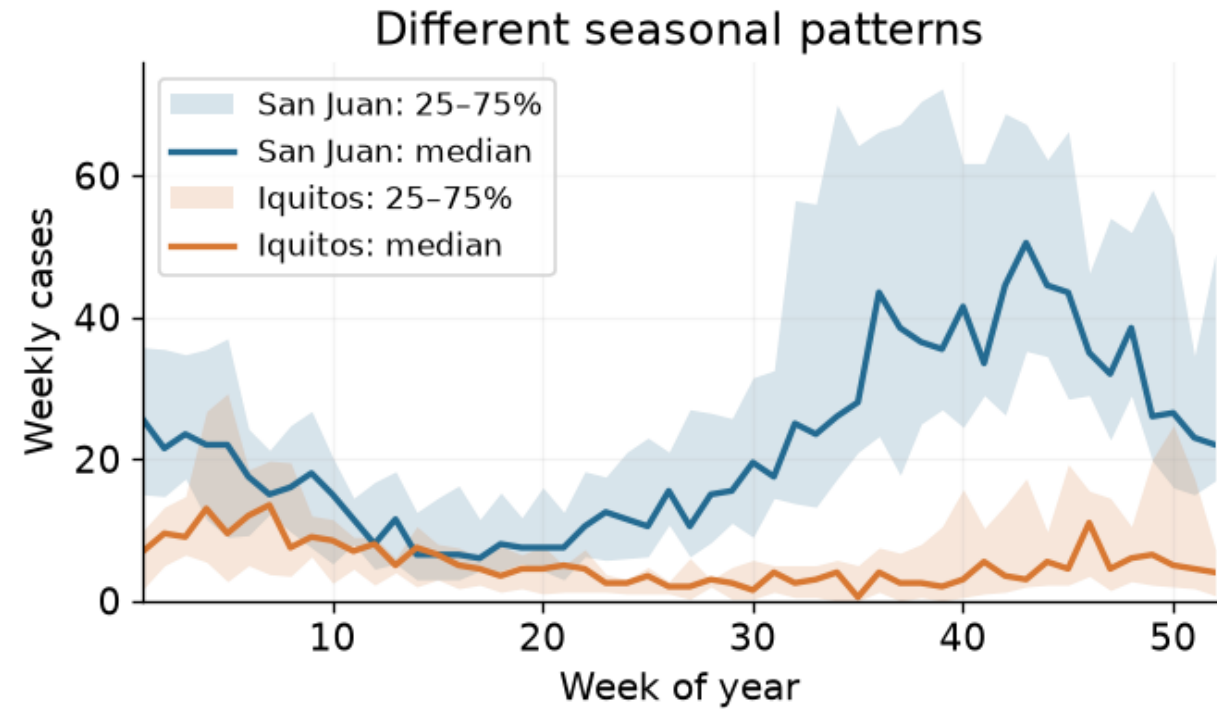
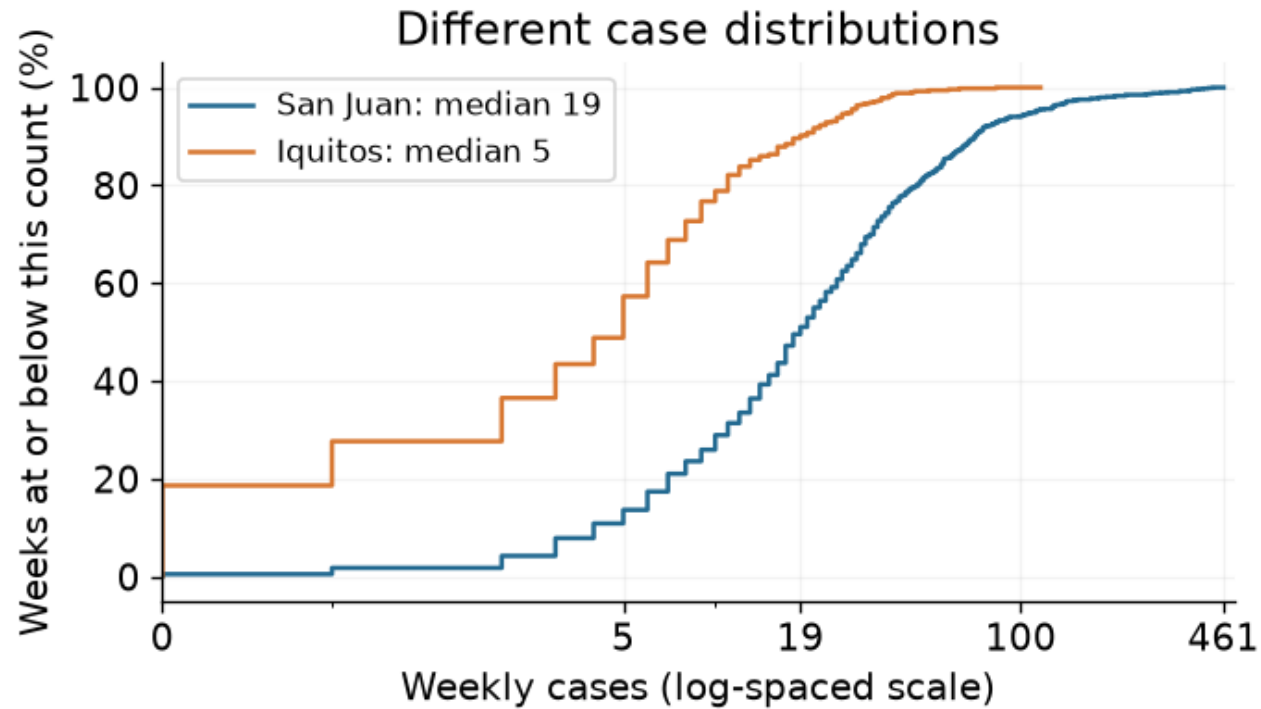
```
test.groupby("city")["week_start_date"]  
    .agg(["count", "min", "max"])
```

Out

city	count	min	max
iq	156	2010-07-02	2013-06-25
sj	260	2008-04-29	2013-04-23

Each city's test period follows its training period: about five years for San Juan and three years for Iquitos. We therefore validate on later weeks.

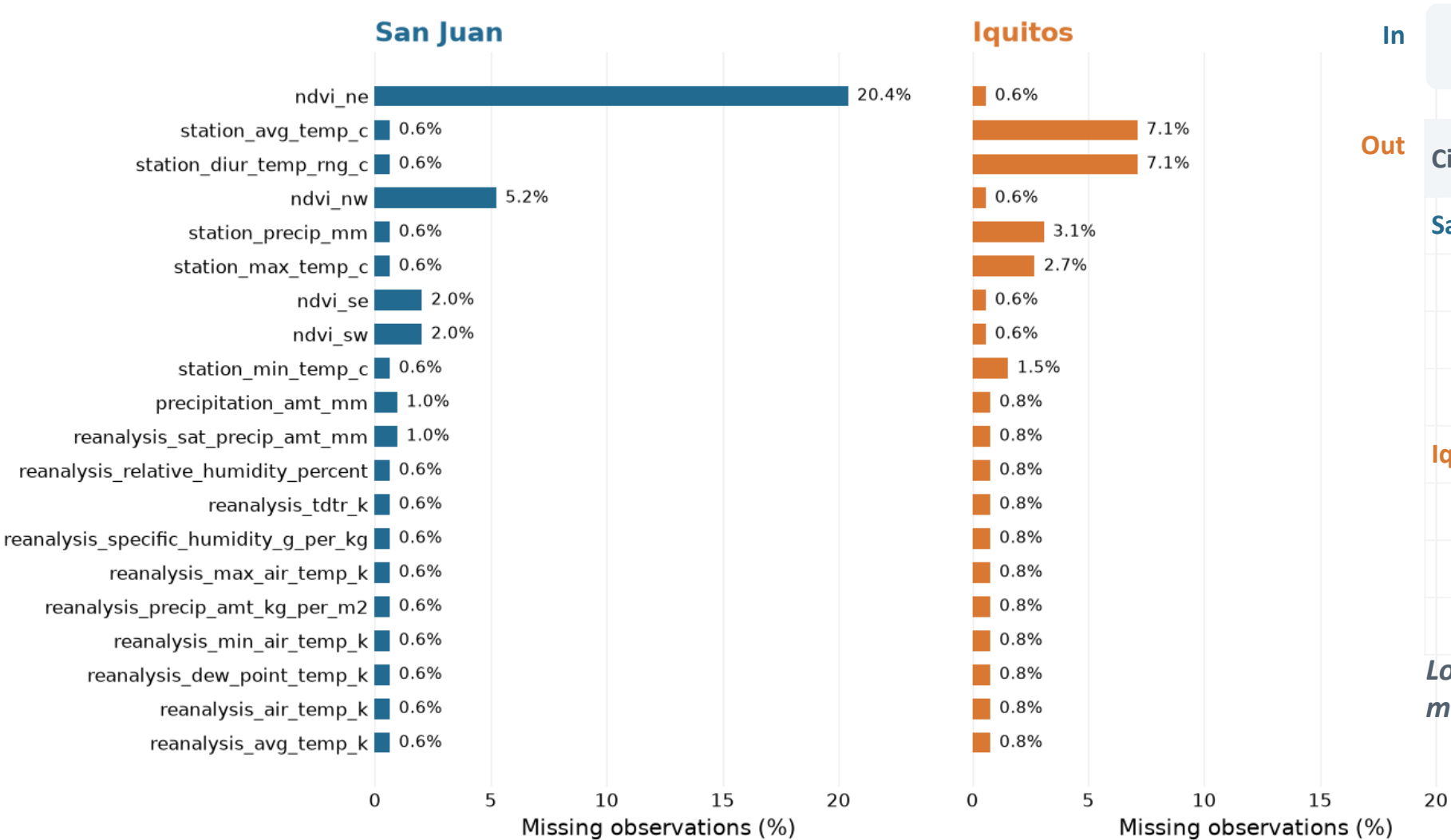
The cities have different case levels and seasons



Decision: perform EDA separately, then prepare and fit a model for each city.

Vegetation has the most missing values

Missing environmental measurements



In

```
ndvi_missing_summary(eda)
```

Out

City	Feature	Missing (%)	Longest gap
San Juan	ndvi_ne	20.4	15
	ndvi_nw	5.2	15
	ndvi_se	2.0	14
	ndvi_sw	2.0	14
Iquitos	ndvi_ne	0.6	1
	ndvi_nw	0.6	1
	ndvi_se	0.6	1
	ndvi_sw	0.6	1

Longest gap = longest run of consecutive missing observations within each city.

How we fill missing measurements

San Juan vegetation

Typical value (median) for the same week of the year, learned from the training data. If unavailable, the overall training median.

Iquitos vegetation

Linear interpolation between the nearest available measurements before and after the gap.

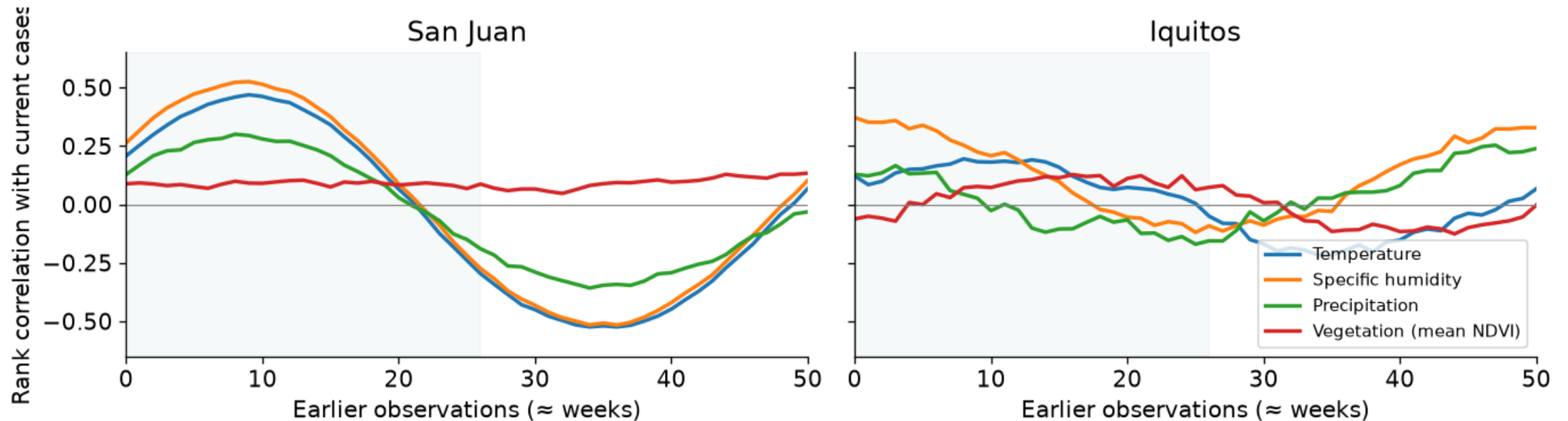
Other weather, both cities

The same interpolation approach for gaps between known values.

Why treat San Juan vegetation differently?

Its gaps run up to 15 consecutive observations. Forward-fill would repeat one old value for months; the same-week median keeps the seasonal shape.

Earlier weather helps explain current cases



The effect is delayed

In San Juan, temperature and humidity correlate most with cases 8–10 weeks earlier

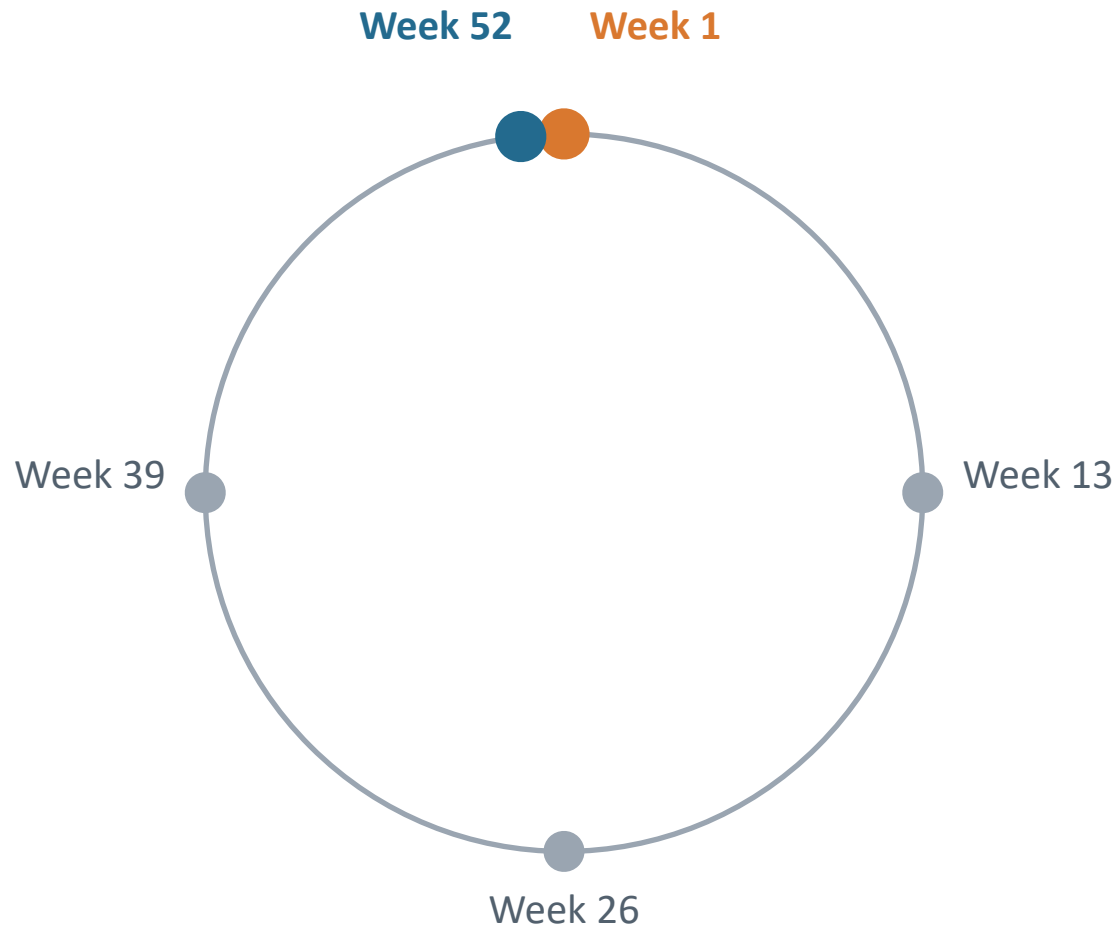
We add lags

Values from 1, 2, 4, 8, 10, 20 and 26 weeks earlier

And rolling summaries

Averages and rainfall totals over the previous 4–26 weeks

Representing the yearly cycle



Weeks 52 and 1 are neighbours, even though their numbers are far apart

We use **cyclical time encoding** for weeks

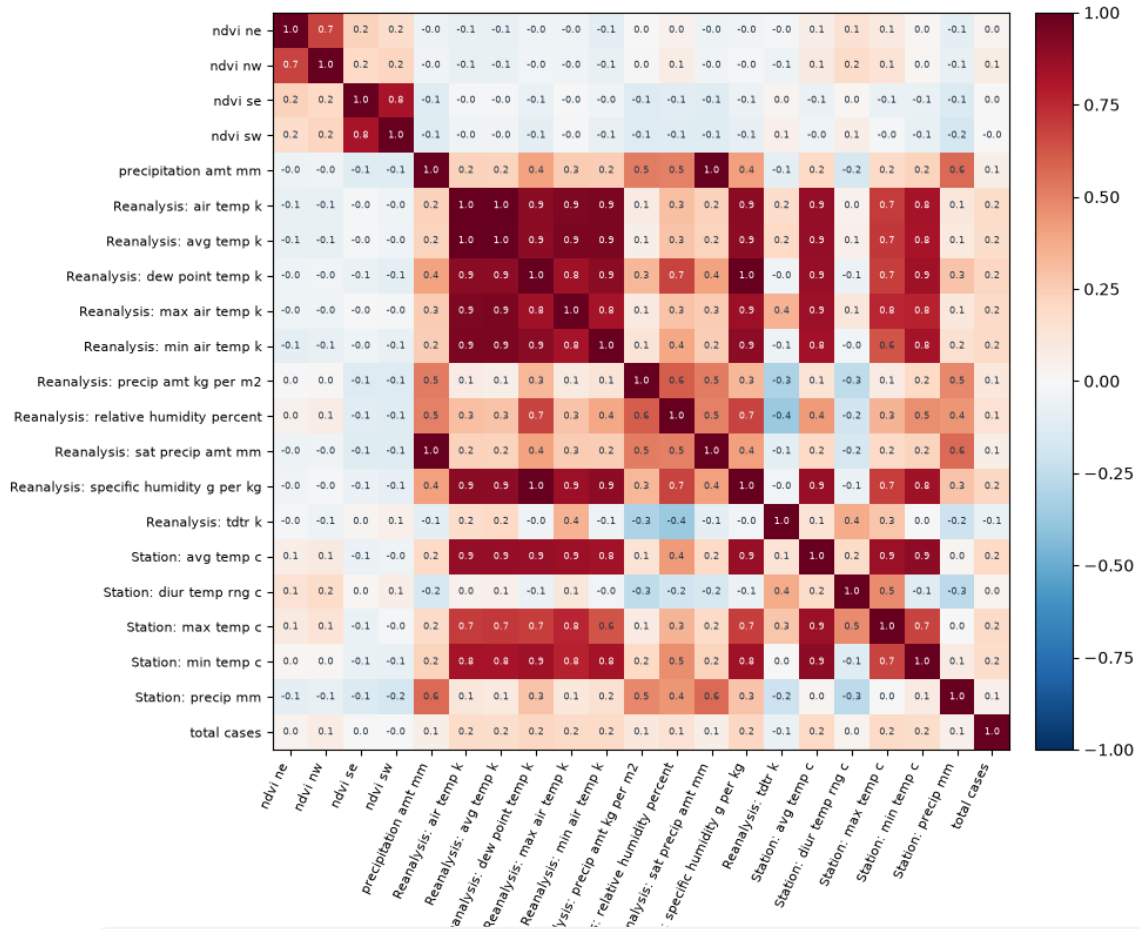
Sine and cosine place each week on a circle

The model sees late December and early January as close together

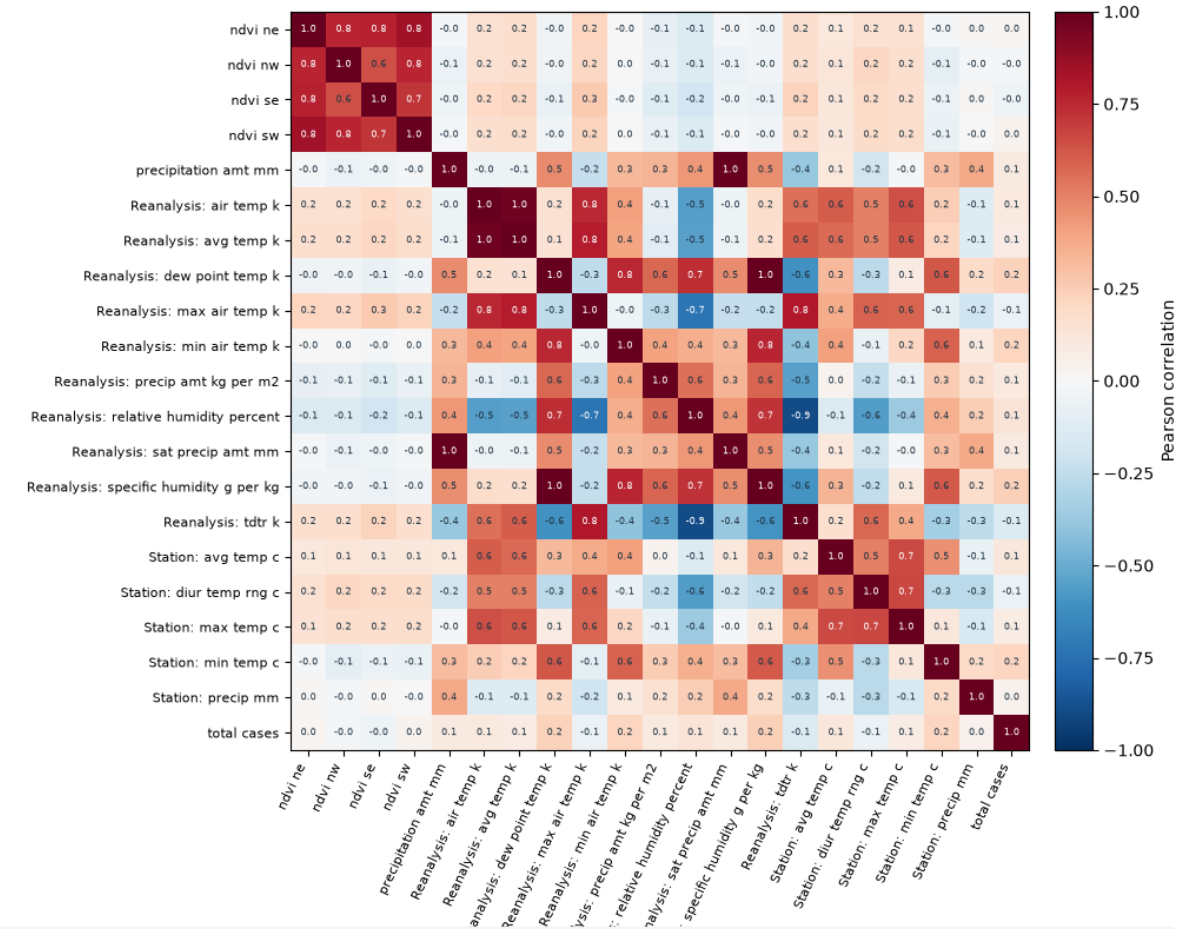
$$\sin(2\pi \cdot \text{week} / 52)$$
$$\cos(2\pi \cdot \text{week} / 52)$$

Checking for redundant inputs

San Juan · environmental features and weekly cases



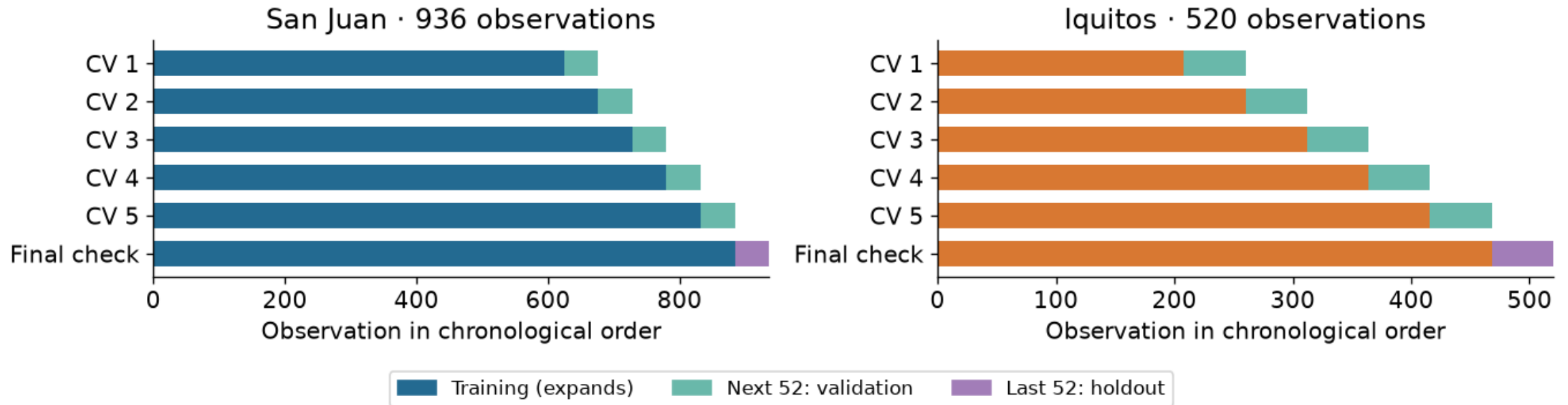
Iquitos · environmental features and weekly cases



Strong correlation suggests overlap, not identity.

Checked directly: **precipitation_amt_mm = reanalysis_sat_precip_amt_mm** in both cities, so the duplicate is **dropped** (along with city, week_start_date and weekofyear).

Time-ordered cross-validation



Keep time order

Train on earlier weeks, validate on later ones

5 folds per city

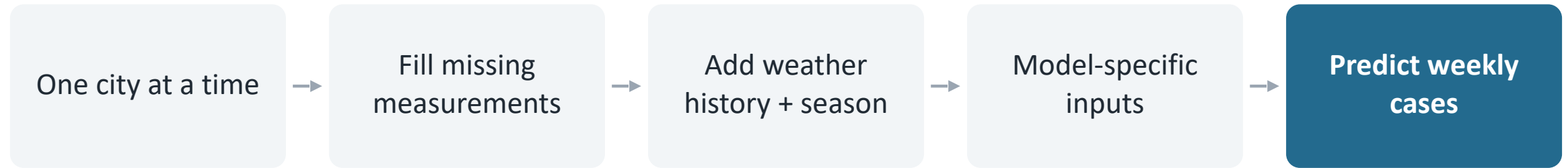
Each predicts the next 52 weeks with more history

Final holdout

Last 52 weeks set aside; choose settings by CV MAE

```
time_cv = TimeSeriesSplit(n_splits=5, test_size=52)
```

One pipeline, refit inside every fold



Avoid leakage

Filling rules, scaling and feature selection are learned from training weeks only

Fair validation

Validation case counts never enter training

Consistent predictions

The same fitted preparation is reused on new data

Feature preparation at a glance

Feature group	San Juan	Iquitos
Vegetation (ndvi_*)	Same-week-of-year training median; overall median as fallback	Interpolate internal gaps
Station, reanalysis and satellite rainfall	Interpolate internal gaps	Interpolate internal gaps
Weather history	Lagged values and rolling summaries	Same
Annual seasonality	Sine/cosine of week of year	Same

DengAI — Predicting Disease Spread

Metrics and Models

October 2026

Metrics

The competition metric is MAE

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

The average number of cases we are off by, per week.

- One number over all **416 test weeks** (260 San Juan + 156 Iquitos)
- Lower is better; a perfect model scores 0
- **San Juan dominates:** 63% of the weeks and about four times the cases

The target is skewed

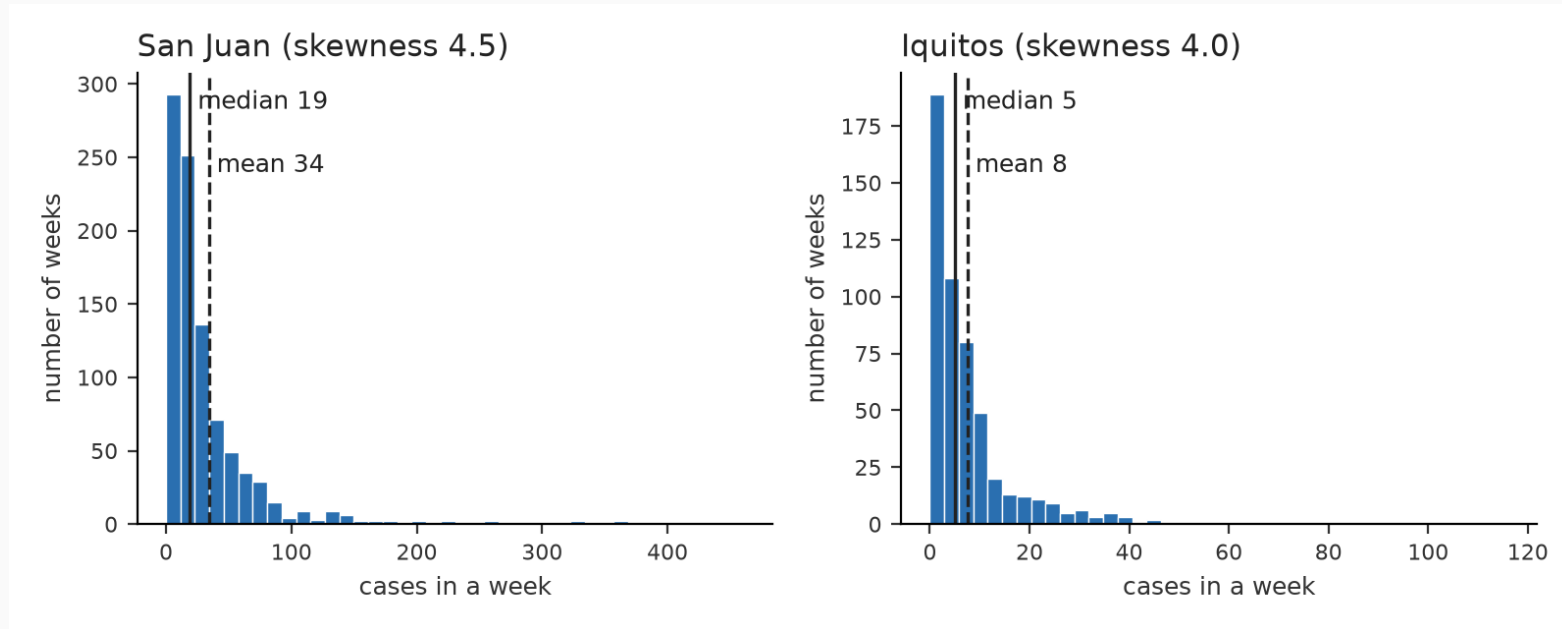


Figure 1: Weekly cases: most weeks are quiet, a few outbreak weeks form a long tail

San Juan: **mean 34, median 19**, 73% of weeks below the mean. Skewness **4.5** (San Juan) and **4.0** (Iquitos).

Why MAE and not RMSE

- RMSE squares errors: one 461-case week counts like **~500 ordinary weeks off by one**
- RMSE is dragged into fitting a few outbreaks; **MAE weights every week equally**
- MAE is minimised by the **median**, RMSE by the **mean** — for skewed counts the median is the robust target
- Consequence: good models here are *conservative*

Models

The pipeline has four branches

`make_city_pipeline` builds one of four shapes, because model families need different preprocessing:

Branch	Steps	Models
tree	impute → history → cyclical → select → model	CatBoost, RandomForest
dense	tree steps + impute + scale	Ridge, Lasso
time_series	impute → history → date + scaled weather	Prophet, SARIMAX
seasonal	calendar columns only	Seasonal median, Shape×level

Why four branches

dense — the lag/rolling warm-up leaves **NaNs** in the first 26 weeks

- Tree models handle NaN natively; RidgeCV raises ValueError
- So linear models get SimpleImputer(median) + StandardScaler

seasonal — the calendar baselines need year and weekofyear, which the tree branch drops

- They **never see the weather**: the baseline any weather model must beat

The nine models

Model	Family	Core assumption
CatBoost	gradient boosting	non-linear effects of lagged weather
XGBoost	gradient boosting	as CatBoost, on RFE-selected features
RandomForest	bagged trees	same, without boosting
Ridge	linear + L2	smooth linear response, all features shrunk
Lasso	linear + L1	linear, but most features are irrelevant
Prophet	additive time series	trend + annual seasonality + regressors
SARIMAX	ARIMA + exog	autocorrelated errors + Fourier seasonality
Seasonal median	calendar only	this week behaves like the same week in past years
Shape × level	calendar only	seasonal <i>shape</i> × typical annual <i>total</i>

Results — holdout MAE

Model	San Juan	Iquitos
CatBoost	15.02	6.27
RandomForest	15.16	8.39
XGBoost + RFE	17.21	3.54
Shape × level	20.72	3.39
Lasso	21.16	4.50
Prophet	21.78	7.16
Seasonal median	23.49	4.41
Ridge	24.46	5.48
SARIMAX	24.56	4.61

Different winners per city — the submission uses **CatBoost** for **San Juan**, **SARIMAX** for **Iquitos**.

What drives the CatBoost predictions?

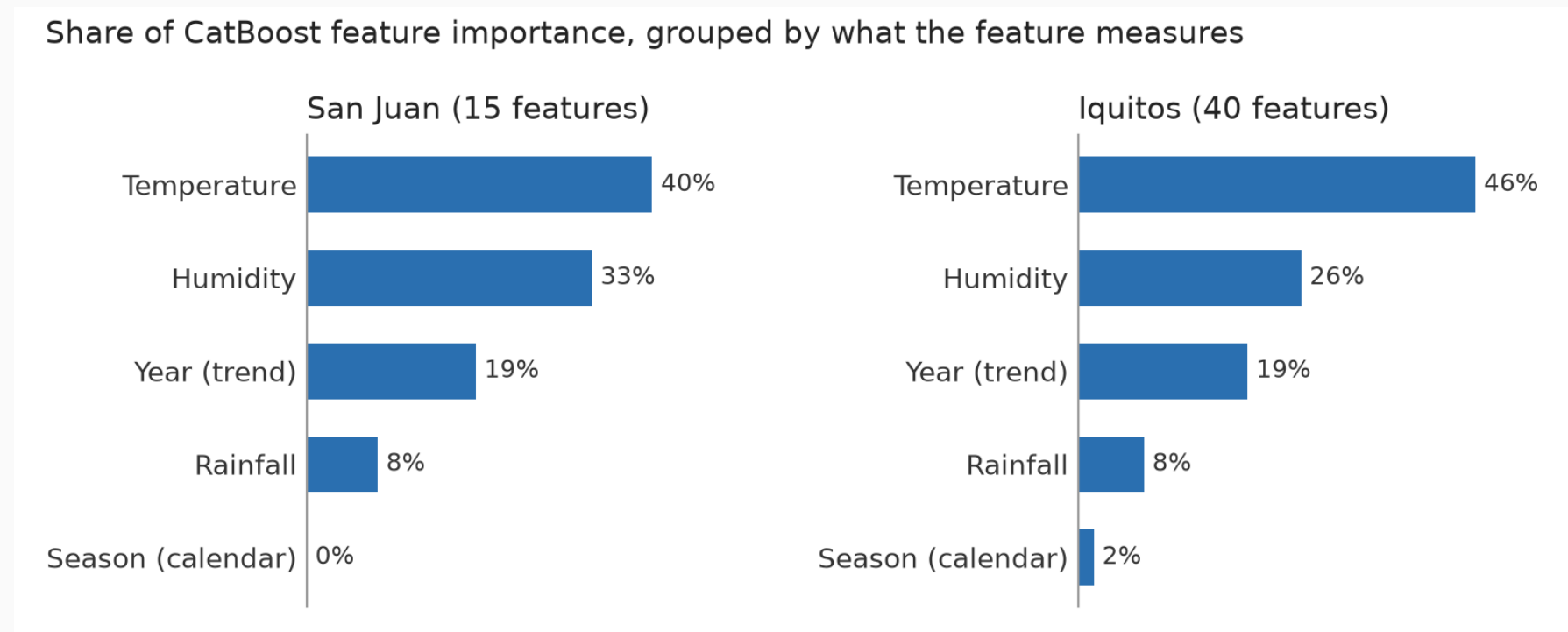


Figure 2: CatBoost feature importance by feature family

- **Temperature and humidity: about 70%** of importance in both cities
- Top features are **8–26-week averages**, not this week's weather
- **year: about 19%** — levels drift between years for non-weather reasons

Why weather is not the whole story

- Climate predicts **when** cases rise within a year, not **how many**
- In Iquitos the calendar-only `Shape × level` (3.39) beats every weather model
- Epidemic size depends on the circulating **serotype** and population immunity — neither is in the data

- **Metrics** — MAE, because the target is skewed (4.5): a few outbreak weeks would dominate RMSE
- **Models** — nine models in four pipeline branches; winners differ by city
- **Feature importance** — months of temperature and humidity; weather says *when*, not *how many*

THAT'S NOT ENOUGH

FORECASTING DENGUE BEYOND THE LEADERBOARD

Why a better score is not enough

From prediction accuracy to decisions that affect patients

WE HAVE TO GO DEEPER

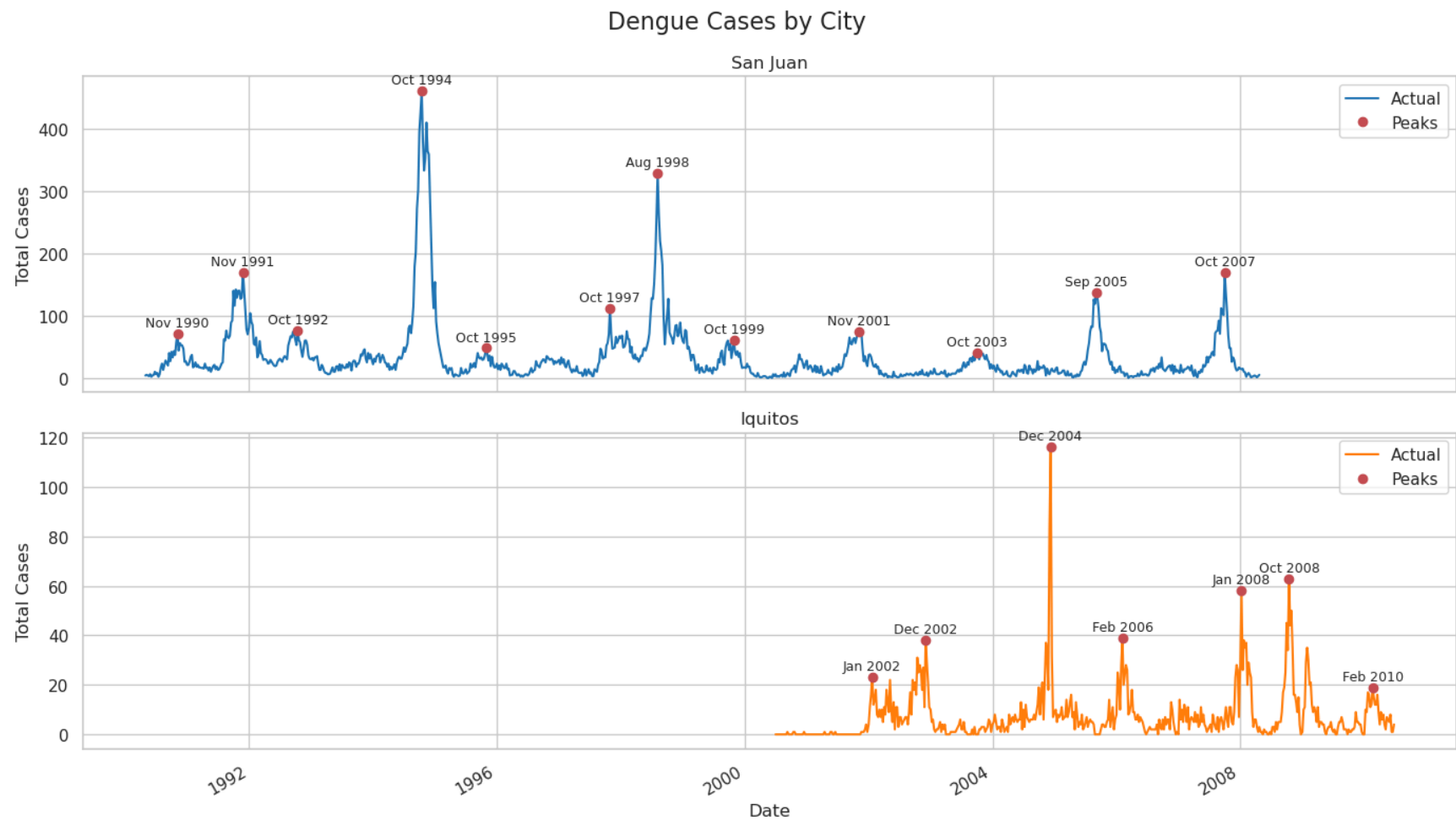
Following the initial solution, proposed by Peter Bull from DrivenData who noticed that variance in the data is much larger than mean that can be modelled via the negative binomial distribution, we can do even better using time series.

CITY	MEAN	VARIANCE
San Juan	34.18055555555556	2640.045439691045
Iquitos	7.565384615384615	115.89552393656412

For example, using SARIMAX for Iquitos, we can elevate the performance on hold out:

	HOLDOUT MAE
sarimax	4.61
catboost	6.27

Time-series based solutions for this problem were heavily explored in the papers, considering this dataset, and proved the efficiency on the leaderboard.



If You're So Smart, How Come You Ain't Rich?

If the problem is so easy, why our GOAT team is not top 1 in the leaderboard?

#825



Cxx_is_the_best_language

1w 2d ago · 14 submissions

23.6010

Because we did not want to stretch the owl over the globe! At some point, something started to feel wrong.



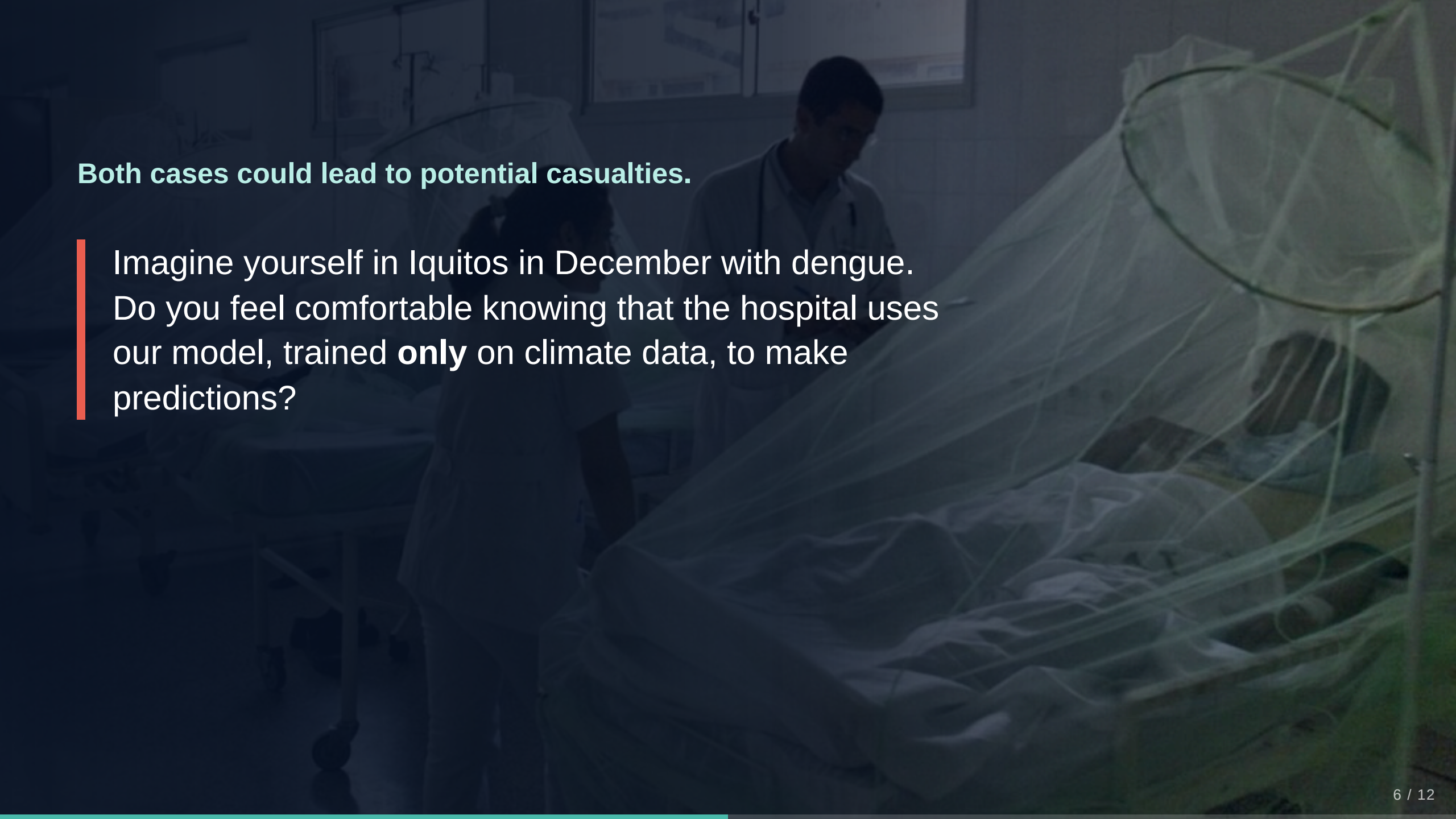
Suppose we want to roll out our model so it can be used by, for instance, the Regional Hospital of Loreto.

Then, our MAE represents how the model's predictions deviation from the true number of cases by an average of X cases per week, encompassing both overpredictions and underpredictions.

Underprediction: The hospital prepares fewer medications than needed, directly risking patient safety and care quality.

Overprediction: The hospital over-allocates resources, leading to expired supplies, financial waste, and strained capacity in other departments.



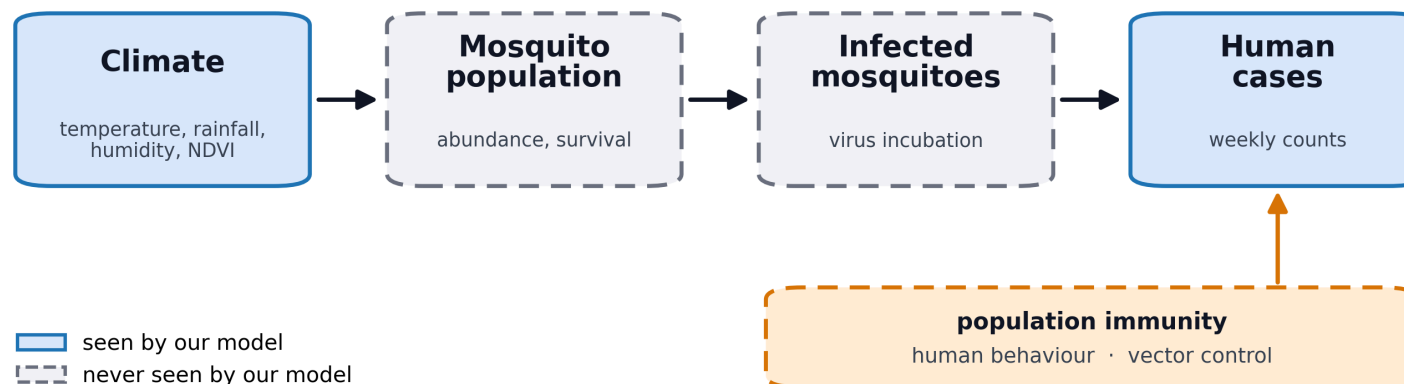


Both cases could lead to potential casualties.

Imagine yourself in Iquitos in December with dengue. Do you feel comfortable knowing that the hospital uses our model, trained **only** on climate data, to make predictions?

What does our model actually see?

Every DengAI feature is a climate or vegetation measurement: temperature, humidity, precipitation, NDVI. Dengue, however, is a chain:



Our model observes the first node and the last one. Everything in between, and everything pushing on the last node, is invisible to it. Even DrivenData's own launch post calls the relationship between dengue and climate complex.

Predicting a curve is not the same as understanding what moves it. A model that has never seen a mosquito cannot tell us what happens when the mosquitoes change.

Findings

Transmission was dominated by single serotypes, first DENV-3 (2001-2007) then DENV-4 (2008-2010). After 2003, incidence fluctuated inter-annually with outbreaks usually occurring between October and April. We detected a strong positive autocorrelation in case counts at a lag of ~70 weeks, indicating a shift in the timing of peak incidence year-to-year. All climatic variables showed modest seasonality and correlated weakly with the number of reported dengue cases across a range of time lags. Cases were reduced after citywide insecticide fumigation if conducted early in the transmission season.

Conclusions

Dengue case counts peaked seasonally despite limited intra-annual variation in climate conditions. Contrary to expectations for this mosquito-borne disease, no climatic variable considered exhibited a strong relationship with transmission. Vector control operations did, however, appear to have a significant impact on transmission some years. Our results indicate that a complicated interplay of factors underlie DENV transmission in contexts such as Iquitos.

Why not just add the mosquitoes?

As far as we could find, there is no public weekly mosquito series for the DengAI period (1990-2010) for San Juan.

- **Iquitos, Peru:** Mosquito-abundance data cover **1999–2010** ([Reiner et al., 2019](#)).
- **San Juan, Puerto Rico:** Mosquito-community data were collected in **2018–2019** ([Yee & Scavo, 2021](#)).
- **Metropolitan San Juan:** Post-hurricane mosquito surveillance began after Hurricanes Irma and Maria in **2017** ([Barrera et al., 2019](#)).
- **San Juan dengue surveillance:** Mosquito data were collected during the **2024 dengue epidemic** ([Otero et al., 2025](#)).
- **Bayamón:** The mosquito-control and surveillance program began in **2024** and is planned to last two to three years ([Puerto Rico Vector Control Unit, n.d.](#)).

Summary: Four of the five sources cover **2017–2024**, while only the Iquitos dataset reaches back to **1999**. This recent concentration likely reflects expanded surveillance following dengue, chikungunya and Zika outbreaks and Hurricane Maria, as well as 2010-2013 dengue outbreak, leaving little comparable data for the 1990s.

Some of the data is accesible only by request and we had no time to email the authors.

The presence of such continued studies serve as an indicator that the experts in the field actually understand the necessity for the stronger correlated data.

Insights from [Barrera et al., 2019](#) say that

Severe storms are expected to increase the number of mosquitoes by increasing or decreasing the edge surface of groundwater habitats, filling cavities/containers with rainwater, and generating additional container aquatic habitats in destroyed or abandoned properties (damaged appliances, inundated floors/roofs, and rubble).

Hence, we also need to account for the movements caused by the natural disasters.

Interesting fact: the highest peak in San Juan was in 1994. Look at what happened during that year on January 7:



Morris J. Berman oil spill, San Juan, January 7, 1994

We could not find any studies exploring the causal link between the oil spill and the dengue outbreak that year.

If you're so smart, how come you ain't rich?

Because a leaderboard rank is a number, whereas a hospital's stock of medicine is a decision that affects people's lives.

Before taking responsibility for a clinical use, we would need:

1. **Mosquito surveillance:** weekly adult trap counts and virus infection rates, for the same years as the cases.
2. **Immunity:** serotype data and serosurveys.
3. **Intervention logs:** when and where vector control happened.
4. **Natural-disaster data:** storm dates, tracks, rainfall and flood extent, and damage to housing.
5. **Quantile forecasts** with checked coverage, instead of a point forecast tuned for MAE.
6. **A test of our own claim:** the same model with and without climate covariates (and with a disaster log added), compared on the epidemic years.

The model is likely to forecast the curve. We cannot yet say that it understands why the curve moves, and in a hospital, this is what matters the most.

Thanks for your attention!

Questions?